

Agastya Teja Atluri

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SUMMARY

Results-driven AI/ML Software Engineer with 4+ years of experience building scalable, high-impact systems across GenAI, MLOps, and backend infrastructure. Proven ability to reduce operational costs, accelerate deployments, and enhance user engagement through production-grade AI/ML pipelines, LLM applications, RAG assistants, multimodal chatbots, fine-tuning, and microservices. Expert in Python, FastAPI, Docker, and cloud-native architecture, with a strong foundation in system design, real-time APIs, and vector-based search systems.

EXPERIENCE

Software Engineer AI/ML – Key Solutions Inc, Fremont, CA

October 2024 - Present

- Architected a production RAG pipeline using LangChain, ChromaDB, GPT-4 that auto-verifies protocol forms (IACUC, IRB, IBC), cutting compliance review time 40 % and serves 15k+ monthly requests.
- Optimized GPT-4-Turbo inference latency 25 % via prompt compression, streaming, and asyncio-based FastAPI back end, benchmarking with custom evaluation harness using Ragas, pytest.
- Fine-tuned LLaMA 2 and Mistral-7B with LoRA on 10 k domain Q&A pairs, slashing hallucinations 28 % and reducing token out-of-window errors.
- Designed TypeScript SDK & React hooks for internal teams to consume AI endpoints, enabling less than 2-day integration into three client applications.
- Championed container-native MLOps like GitLab CI, Docker, AWS ECS blue-green deployments with zero downtime across 3 environments.

Software Engineer – Antela Group, Atlanta, GA

January 2024 - October 2024

- Built embeddings-driven analytics dashboard using Python and Next.js with anomaly detection and NLP insights, exec decision cycle speed up 40 %.
- Stood up Kafka to Spark streaming ETL ingesting 100k msgs/day into PostgreSQL & pgvector for real-time recommendations.
- Deployed PyTorch models via MLflow and AWS SageMaker, automating evaluation and rollback on drift, improving model reliability 35 %.

Graduate Research Assistant – Georgia State University, Atlanta, GA

August 2022 - December 2023

- Predicted call center load for Southern Company using Python, ARIMA, and XGBoost, cutting caller wait time 20 % and reduced abandoned calls 30 %.
- Built Airflow orchestrated ETL pipelines using SQL, Pandas that ingested 10+ external data streams nightly, shrinking data-prep effort 50 %.
- Deployed Transformer based sentiment chatbot using PyTorch, Hugging Face to triage tickets, raising agent productivity 25 %.
- Automated Tableau dashboards via GitLab CI/CD for live KPI & sentiment tracking, eliminating manual reports and boosting refresh cadence 12 times.

Software Engineer – Deloitte, Hyderabad, India

August 2021 - August 2022

- Developed REST microservices using Java, Node.js for Oil & Gas client's digital-commerce rollout, increasing customer engagement 25 %.
- Designed SAP Hybris and Azure SQL data layer, cutting query latency 50 % and driving 60 % faster data retrieval for downstream services.
- Implemented GitLab CI/CD pipelines with blue-green deployments, tripling release frequency and slashing rollback time 70 %.
- Created Postman test suite & Grafana monitoring, achieving 99.9 % API uptime and zero critical incidents over 6 months.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, Javascript, C, C++, HTML, CSS

AI Techniques: RAG, embeddings (OpenAI, BGE), LoRA fine-tuning, agentic orchestration, prompt engineering, evaluation with LangSmith, Ragas

Web & API Development: FastAPI, Flask, RESTful APIs, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Node.js, React, Next.js, Tailwind CSS

Infrastructure & DevOps: AWS, Google Cloud Platform (GCP), Docker, Kubernetes, Kafka, GitLab CI/CD, Airflow, Spark, Tableau, Power BI, Microsoft Excel

Databases: PostgreSQL, ChromaDB, pgvector, FAISS, MySQL, MongoDB, Microsoft SQL Server, Redis

PROJECTS

Deep Hybrid Recommender System | PyTorch, FastAPI, Redis, Docker, PostgreSQL, AWS EC2

Built a hybrid recommender system combining collaborative filtering, content-based models, and neural networks. Deployed as a containerized microservice with real-time Redis caching and RESTful APIs, achieving 35% improvement in personalized content ranking precision on test datasets.

Customer Churn Prediction | EC2, S3, RDS, Lambda, SageMaker, Flask, TensorFlow, PyTorch

Devised a bank customer churn prediction model using various Machine learning algorithms such as Decision Tree, Random Forest, AdaBoost, Hist Gradient Boosting, XGBoost, and SVM resulting in an overall accuracy of 90%.

Domain-Tuned Medical Assistant | LLaMA 2, LoRA, peft, transformers, accelerate, Gradio, Docker, Google Colab

Fine-tuned LLaMA 2 on domain-specific medical Q&A data using Low-Rank Adaptation (LoRA) with Hugging Face's peft and transformers libraries, optimizing for GPU efficiency and reduced hallucinations; deployed via Gradio with evaluation benchmarks showing 28% improvement in factual accuracy on validation set.

FluxPilot | Python, LangChain, MLflow, PyTorch, Hugging Face Transformers, FastAPI, Docker, Prefect, AWS S3, SageMaker, GitHub Actions

Engineered FluxPilot, an end-to-end MLOps pipeline that continuously fine-tunes domain LLMs. Combined LangChain prompt modules, Prefect-driven retraining, MLflow and SageMaker for tracked, scalable deployments behind versioned FastAPI endpoints, enabling real-time monitoring and boosting task accuracy 60 %.

EDUCATION

Georgia State University

August 2022 - December 2023

Masters in Data Science and Analytics

Atlanta, GA

VNR Vignana Jyothi Institute of Engineering and Technology

July 2017 - July 2021

Bachelors of Technology in Electronics and Communication Engineering

Hyderabad, India